

Simultaneous equations



UNIT 5: Adding or subtracting to eliminate a variable

QUESTION 1 Solve simultaneously after adding the two equations together.

a $a + b = 11$
 $a - b = 3$

b $2x + y = 17$
 $x - y = 1$

c $3m - n = 22$
 $2m + n = 23$

d $5p + 2q - 26 = 0$
 $3p - 2q - 6 = 0$

e $7x + 4y = 113$
 $5x - 4y = 19$

f $6k + 7d = 16$
 $3k - 7d = 29$

QUESTION 2 Solve simultaneously, after subtracting one equation from the other.

a $5x + 2y = 39$
 $4x + 2y = 32$

b $9p + q = 79$
 $5p + q = 47$

c $6a + 5b = 62$
 $4a + 5b = 38$

d $3a - b = 17$
 $a - b = 3$

e $8m - 3n = 102$
 $5m - 3n = 57$

f $12x - 7y = 94$
 $x - 7y = -5$

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UNIT 6: Solving by elimination

QUESTION 1 Multiply the second equation so that the coefficients of y are the same and then solve simultaneously.

a $3x + 2y = 21$
 $x - y = 2$

b $5x + 8y = 64$
 $3x + 2y = 30$

c $7x - 6y + 5 = 0$
 $4x - 3y + 2 = 0$

QUESTION 2 Solve simultaneously, using the elimination method after multiplying one or both equations.

a $5a - 4b = 9$
 $3a + b = 2$

b $6x + y = 25$
 $2x + 3y = 27$

c $3x + 2y = -1$
 $x - 4y = -33$

d $3m + 2n = 10$
 $5m + 3n = 17$

e $9a - 4b = 2$
 $7a - 3b = 1$

f $a + 5b = 18$
 $6a - 2b = 44$

g $2x + 7y - 1 = 0$
 $5x - 3y - 64 = 0$

h $8y + 3z = 104$
 $5y - 6z = 2$

i $m - 6n = 23$
 $8m - 9n = 28$

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UNIT 7: The method of elimination

QUESTION 1 Solve the following pairs of simultaneous equations by the method of elimination.

a $x + 2y = 8$
 $3x - 2y = 4$

b $2x - 3y = 6$
 $x + 3y = 9$

c $2x + 5y = 19$
 $3x - 5y = 6$

d $2x + y = 10$
 $3x - y = 5$

e $3x + 4y = 14$
 $x + y = 3$

f $2x - y = 3$
 $x - 2y = 9$

QUESTION 2 Use the method of elimination to solve the following pairs of simultaneous equations.

a $3x + 4y = 12$
 $x - y = 4$

b $4x + 5y = 22$
 $x + y = 10$

c $2x + 3y = 11$
 $x - y = -2$

d $5x - 3y = 9$
 $3x + y = 4$
